Vertex Form of a Quadratic Function

Name _____

Algebra 2 Lesson 2-1 (fast track) Date _____ Period _____

Learning goal: I can find the vertex, axis of symmetry, maximum or minimum, domain and range of a quadratic function in vertex form, and write its equation from the vertex and one other point.

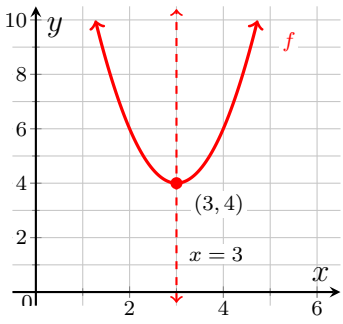
Language goal: I can explain how vertex form describes a parabola using the words vertex, axis of symmetry, maximum, minimum and opens upward or downward.

On the quiz: You will (1) tell how $g(x) = (x+2)^2+4$ moves $y = x^2$ (full Savvas quiz only), (2) find a parabola's highest point from vertex form (Example 2), (3) write vertex form from the vertex and a point (Example 4), (4) read the vertex and y-intercept off a graph, find a, write $y = ax^2 + bx + c$ (Example 4), (5) write a shifted parabola as $y = ax^2 + bx + c$ (full Savvas quiz only).

Example 2: Determine Key Features of a Quadratic Function

Vertex form: $y = a(x-h)^2 + k$, $a \neq 0$. The **vertex** is (h, k); the **axis of symmetry** is the line $x = h$. If $a > 0$ the parabola **opens upward** and k is the **minimum**; if $a < 0$ it **opens downward** and k is the **maximum**. The domain is always all real numbers.

What are the key features of the quadratic function $f(x) = 2(x-3)^2 + 4$?



1. Match it to $a(x-h)^2 + k$: $a = 2$, $h = 3$, $k = 4$.
2. **Vertex:** $(h, k) = (3, 4)$. **Axis of symmetry:** $x = 3$.
3. The 2 indicates that the graph **opens upward** ($2 > 0$) and is vertically stretched ($|2| > 1$).
4. Opens upward, so the vertex is the lowest point: **minimum** 4.
5. **Range:** $\{y \mid y \geq 4\}$. There are no restrictions on the value of x, so the **domain** is all real numbers.

Say it: The vertex of f is _____, so the axis of symmetry is _____. The parabola opens _____ because $a =$ _____ is _____, so 4 is the _____ value of f.

Try It 2: Key features from vertex form

Identify the vertex, axis of symmetry, minimum or maximum, domain, and range of the function $f(x) = -(x+4)^2 - 5$.

Hint: $x + 4 = x - (____)$, so $h =$ ____.

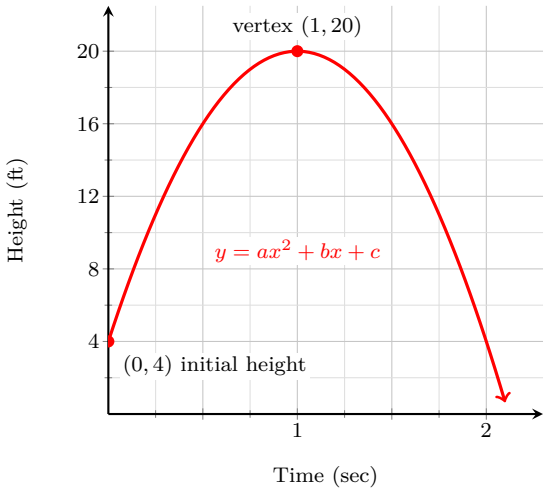
$a =$		opens upward or downward?	
vertex (h, k)		axis of symmetry	
minimum or maximum?		value	
domain		range	

Example 4: Write an Equation of a Parabola Given the Graph

The height of a hit ball is a quadratic function of the time it has been in the air. The ball is hit with an initial height of 4 feet and after 1 second reaches its vertex at 20 feet. What is an equation that defines this function? Write the quadratic equation in vertex form and in the form $y = ax^2 + bx + c$.

1. The **vertex** is $(h, k) = (1, 20)$, so $y = a(x - 1)^2 + 20$.
2. The initial height is the y-intercept, so $(x, y) = (0, 4)$ is on the graph. Substitute it and solve for a: $4 = a(0 - 1)^2 + 20$, so $4 = a + 20$, $a = -16$
3. Vertex form: $\boxed{y = -16(x - 1)^2 + 20}$
4. Expand to get $y = ax^2 + bx + c$: $y = -16(x^2 - 2x + 1) + 20 = -16x^2 + 32x - 16 + 20$ $\boxed{y = -16x^2 + 32x + 4}$

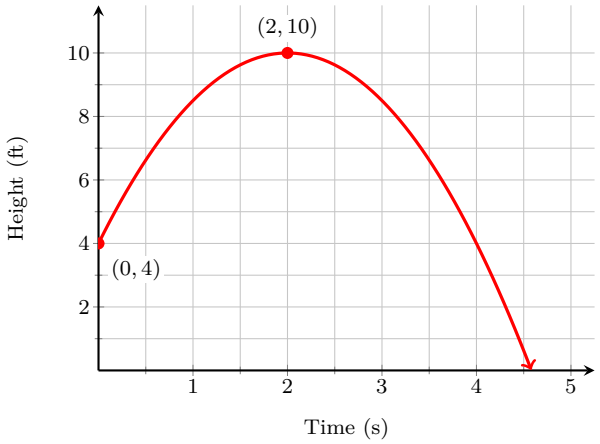
Check: $c = 4$ is the initial height, and $a = -16 < 0$, so the parabola opens downward: the vertex is a maximum.



Say it: The vertex is _____, so $h =$ ____ and $k =$ ____. I substituted the point _____ for x and y and got $a =$ _____. The parabola opens _____ because a is _____, so the vertex is a _____.

Try It 4: Write the equation from the graph

The graph shows the height of a ball with respect to time. What is the equation of the function? Write the equation in vertex form. Then write the equation in the form $y = ax^2 + bx + c$.



Vertex $(h, k) =$ _____
Point $(x, y) =$ _____
$y = a(x -$ ____$)^2 +$ ____
Substitute the point and solve for a:

_____

_____

_____

Vertex form: $y =$ _____

Expand to get the form $y = ax^2 + bx + c$:

_____

_____

$y =$ _____ Which number in this form is the initial height? _____

IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
L8C ★	Characteristics of quadratic functions: equations	Quiz 2: vertex, maximum or minimum from $a(x-h)^2+k$ (Example 2)
URV ★	Write a quadratic function in vertex form from its vertex and another point	Quiz 3 and 4: vertex + one point gives a (Example 4)
KQL	Transformations of quadratic functions	Optional: Quiz 1 and 5 (full Savvas quiz only)
WMS	Characteristics of quadratic functions: graphs	Optional warm-up: key features read off a drawn parabola

Exit question: Amaya's serve (Practice 15)

Amaya is standing 30 ft from a volleyball net. The net is 8 ft high. Amaya serves the ball. The path of the ball is modeled by the equation $y = -0.02(x-18)^2+12$, where x is the ball's horizontal distance in feet from Amaya's position and y is the distance in feet from the ground to the ball.

a. How far away horizontally is the ball from Amaya when it is at its maximum height? Explain.

The vertex is _____. The parabola opens _____ because $a =$ _____, so the vertex is the _____. The ball is _____ ft from Amaya because

b. Describe how you would find the ball's height when it crosses the net at $x = 30$.

Before the quiz, finish these sentences.

In $y = a(x-h)^2+k$, the vertex is _____ and the axis of symmetry is the line _____.

When a is negative, the parabola opens _____ and k is the _____ value.

To find a, I substitute the _____ for h and k and one other _____ for x and y.